

Differential Gearbox

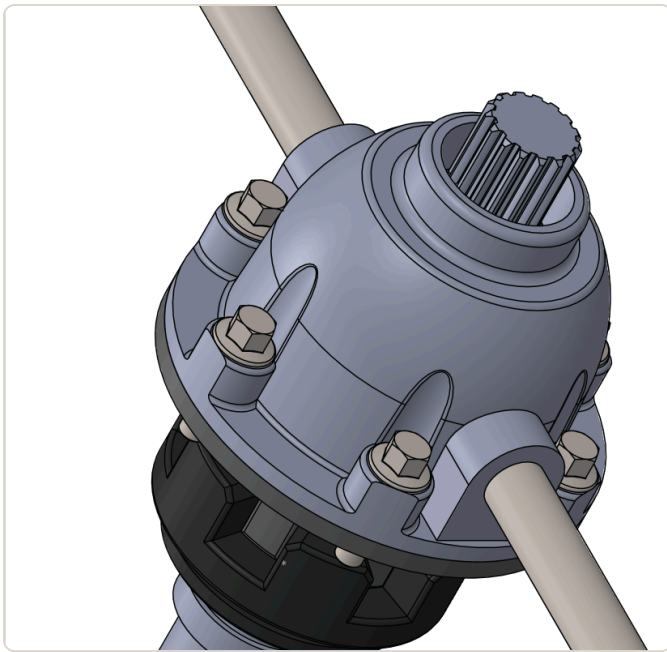
MECHANICAL DESIGN AND CAD PROJECT

Rakin Ahmed • SOLIDWORKS • Assembly, exploded, and detailed orthographic drawings

OVERVIEW

This project is a complete computer aided design of an automotive differential gearbox, the mechanism that transmits engine torque to the drive wheels while allowing them to rotate at different speeds through a corner. The goal was to model a fully defined, manufacturable assembly and to produce the drawing set a shop would need to build it.

The differential converts the rotation of the drive shaft, carried by the crown wheel, into rotation of the two output half shafts. A set of spider (planet) and side (sun) bevel gears inside the carrier splits the torque and accommodates the speed difference between the two wheels during cornering.



Assembled differential, isometric render



Exploded view of the gear train, carrier, and shafts

COMPONENTS MODELED

- Crown (ring) wheel and drive pinion, with matched bevel tooth geometry
- Spider (planet) gears on the cross pin, and the two side (sun) gears on the half shafts
- Differential carrier housing, bearing seats, and fasteners
- Input and output shafts with splined interfaces

APPROACH AND DELIVERABLES

Each component was modeled as a fully constrained SOLIDWORKS part, then brought together in a top down assembly with mates that reproduce the real kinematics of the gear train. The bevel gears were defined so that the pitch cones meet correctly and the tooth counts set the intended torque split.

- Component level part models for every element of the gear train and housing
- A complete assembly with realistic mates and motion relationships
- An exploded view communicating the assembly order
- Dimensioned orthographic general assembly drawings to drafting standard

The project demonstrates mechanical design fundamentals end to end, from gear and shaft design through assembly modeling to a production ready drawing set.